Contact

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Top Skills

Agent-based Modeling
Artificial Intelligence (AI)
Defect Identification

Afshar Shamsi

PHD Candidate @ Concordia University
Montreal, Quebec, Canada

Summary

I design and develop AI systems that bridge research and real-world applications.I’m a PhD candidate at Concordia University (near completion) with 5+ years of experience across machine learning research and applied system development. My work focuses on Vision-Language Models (VLMs), Large Language Models (LLMs), and robust machine learning under distribution shift.I have hands-on experience developing and experimenting with ML systems across the full pipeline, from model training and adaptation to designing and prototyping scalable workflows. This includes working with multimodal models that combine visual understanding and language reasoning, as well as developing LLM-based pipelines involving retrieval, tool interaction, and structured outputs.I’m particularly interested in advancing AI systems that unify perception, reasoning, and adaptability in real-world environments.Open to opportunities in Applied AI, Agentic AI, and AI Research & Development, with a focus on multimodal and robust systems.

Experience

Novatek International
Machine Learning Researcher
September 2025 - Present (9 months)

Develop AI models for analyzing image and video streams captured with 2D cameras to detect different poses. Implement fall detection algorithms that leverages pose analysis to accurately identify incidents in real time.

Concordia University
Research Assistant
January 2023 - Present (3 years 5 months)

UNSW Graduate School of Biomedical Engineering
Research Assistant
October 2024 - December 2024 (3 months)

Focusing on online test-time adaptation techniques to improve keypoint estimation for pose estimation, addressing cross-domain shifts and ensuring robust performance across diverse environments.

Geo-SaS (Sydney, Australia)
AI Developer
February 2024 - September 2024 (8 months)

Developing an AI model for multi-class object detection in boreholes, designed for mining industries. The model aims to accurately detect key classes such as joints, crush zones, boxes, and empty spaces, while also providing critical metrics like Rock Quality Designation (RQD), Joint Number (Jn), and other relevant geological features ("www.icore.org.au").

Ortho BioMed Inc. (ON, Canada)
Machine Learning Researcher
January 2023 - September 2023 (9 months)

Survival analysis, Kidney transplant, Healthcare

UNSW BioMedical Machine Learning Lab (BML)
Machine Learning Researcher
April 2022 - October 2022 (7 months)

Improving the trustworthiness of AI models for Healthcare

Commercial Production Trading Asoubar Company (ASR)
Electrical Engineer
April 2015 - March 2021 (6 years)

Education

Concordia University
Doctor of Philosophy, Information Technology · (January 2023 - April 2026)

University of Tabriz
Master of Science, Electrical and Control engineer · (2011 - 2013)

University of Tabriz
Bachelor of Science, Electrical and Electronics Engineering · (2009)